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HW 4 Lab Report: Nitrile vs Vinyl Mechanical Test

Introduction

Biomedical engineering plays a crucial role in developing materials that meet the exact demands of healthcare applications. Understanding the mechanical properties of materials used in medical devices and protective equipment is essential for ensuring their effectiveness and safety. In this context, our experiment focused on analyzing the engineering stress and strain in a cut-out section between a nitrile and vinyl glove. Our hypothesis centers on the distinctive mechanical behaviors exhibited by glove samples when subjected to an applied load or force. Our hypothesis assumes that nitrile, recognized for its robust and rigid molecular structure, will result in a higher Young's Modulus when compared to vinyl. Conversely, given vinyl's inherent flexibility, we anticipate a greater engineering strain, as this material is more prone to deformation compared to the less-yielding nature of a rigid material like nitrile, which is expected to exhibit a lower engineering strain.

Methods

We experimented to analyze the mechanical properties of the nitrile glove sample under an applied load. The experiment utilized a standard tensile testing machine to apply a controlled force to the glove specimens.

To prepare, we cut out sections of the nitrile glove sample, while another group does the vinyl material. the glove specimen was securely clamped into the grips of the tensile testing machine. The initial stretch of the material was measured without any additional weight. A gradual and controlled force was applied by adding ball bearings in a canopy (balloon rig) and stretching the specimens until failure occurred. We measured the amount of ball bearings in the canopy at these values: 1, 2, 4, 6, 8, 10, 15, 20, 25. Those amounts were translated to the amount of downward force exerted in Newtons.

Engineering Stress:

To find engineering stress, we used the measured width and thickness to estimate deformed area, then divided force by area to obtain engineering stress. The assumption is that the area does not change, which may or may not be accurate for the chosen material.

$$\sigma = \frac{Force}{Deformed\ Area}$$

Engineering Strain:

Engineering strain is the relative change in sample length.

$$\varepsilon = \frac{Change\ in\ Lenght}{Original\ Lenght}$$

Young's Modulus:

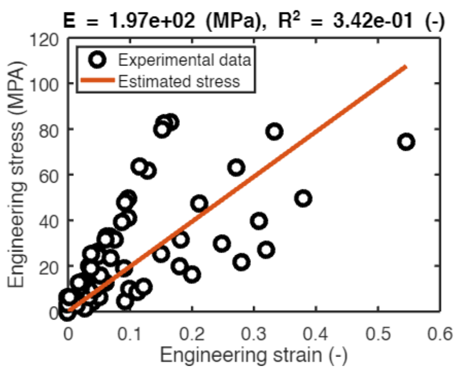
Young's Modulus is linear regression. We can also use a polynomial curve fitting function in MATLAB to find it.

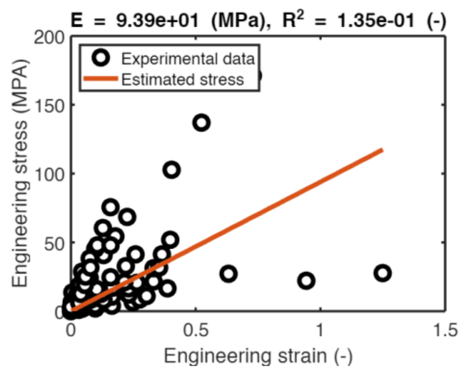
$$E = \frac{\sigma}{\varepsilon}$$

In conducting this experiment, an assumption we made is assumed the area of our chosen object to remain constant throughout the stretch when calculating the engineering stress.

Results

Below is my group's plot of the experimental stress-strain data. Also provided are our Young's Modulus value and R^2 value. **Vinyl** is on the left and **Nitrile** is on the right. *[Note: For our group, measurement starts at 4 ball bearings due to time constraints and nitrile specimen dislodged from top clamp at 14 ball bearings).* Throughout the test, we recorded load and displacement data to capture the complete stress-strain curve.





The R^2 values for both graphs were notably low, with vinyl at $R^2 = 0.342$ and the nitrile sample at $R^2 = 0.135$. Consequently, both samples encountered challenges in achieving a robust linear fit and demonstrated a weak correlation with linear regression. Another significant distinction between the two samples was observed in their engineering strain. Nitrile exhibited a considerably higher engineering strain, approaching 1.3, whereas vinyl had an engineering strain of approximately 0.55.

Discussion

The experiment's outcomes failed to validate the hypothesis, as vinyl exhibited a higher Young's Modulus of 187 MPa in contrast to nitrile's 93.9 MPa. The accuracy of Young's Modulus estimate is questionable, given the poor linear fit observed in both graphs, where the reliability of Young's Modulus is directly linked to the slope. Additionally, the data contradicted expectations for engineering strain, as it was anticipated that vinyl would have a significantly higher engineering strain than nitrile. However, nitrile surpassed vinyl's engineering strain by more than double. These unexpected results could be attributed to various factors, such as the use of a flaw in the tester and the presence of human errors. Notably, instances of sample dislodgment from the tester occurred in the nitrile sheet, skewing the data. These errors contributed to a lower R^2 value, indicating a poor fit, and consequently, yielding unreliable test results.

Now, if we were given a budget of $2000 to enhance our mechanical testing device, What I would change is the data collection method. The addition of real-time data logging and analysis capabilities would enhance the efficiency of the testing process. This could involve integrating advanced software that allows for immediate visualization and interpretation of results during the experiment. Also, upgrading the grips and fixtures would be beneficial as our glove subject dislodged 14 ball bearings, possibly disrupting the data.